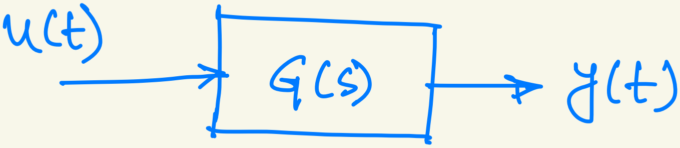
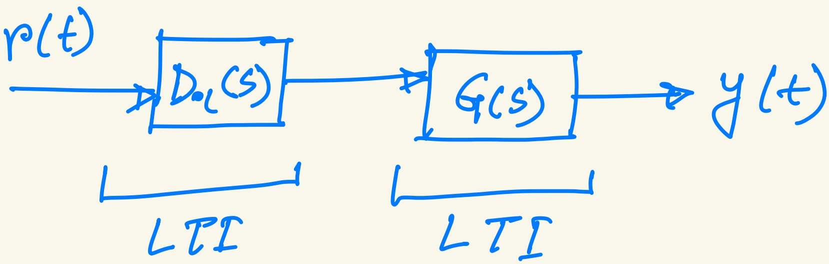
Lecture 23



Our goal is to follow a reference signal $r(t)$ or we want to turn an unstable system into a stable one.



$$\frac{Y(s)}{R(s)} = H_{ol}(s) = D_{ol}(s)\,G(s)$$

Can we cancel poles on the riht-half plane?

$$G(s) = \frac{1}{ml^2} \cdot \frac{1}{s^2 + b/ml^2 s - g/L}$$

$$ml^2 = 1$$

$$g/L = 1$$

$$b = 0$$

$$G(s) = \frac{1}{s^2 - 1}$$

$$= \frac{1}{(s+1)(s-1)}$$

$$D_{ol}(s) = s - 1$$

$$H_{ol}(s) = s-1 \cdot \frac{1}{(s+1)(s-1)}$$

$$= \frac{1}{s+1}$$

Due to noise $g/L = 0.9999$

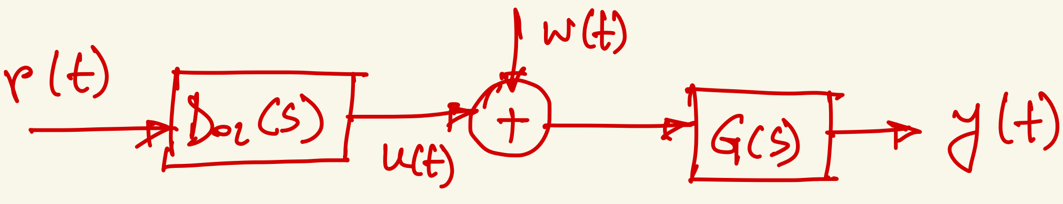
$$H_{ol}(s) = D_{ol}(s)\, G(s)$$

$$= \frac{s-1}{\left(s+\sqrt{0.9999}\right)\left(s-\sqrt{0.9999}\right)}$$

$$= \frac{A}{s+\sqrt{0.9999}} + \frac{B}{s-\sqrt{0.9999}}$$

So, the unstable pole does not get canceled.

In general systems are subject to disturbances $w(t)$



$$Y(s) = D_{ol}(s)\,G(s)\,R(s) + \underbrace{G(s)\,W(s)}$$

We can see that G(s) is getting influenced by noise. And the open loop system does not have any mechanism to mitigate noise.